

EXPLORER

TASK1_DATACOLLECTION

Data.CSV

Task1.py

Task2.py

Task3.py

Task1.py

```
1 import pandas as pd
2 df=pd.read_csv("Data.CSV")
3 print("Dataset Loaded successfully")
4 print(df)
5
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task1.py

Dataset Loaded successfully

	Name	Marks
0	Arun	85
1	Kumar	90
2	Priya	78
3	Divya	88

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

powershell

powershell

> OUTLINE

> TIMELINE

EXPLORER

TASK1_DATACOLLECTION

Data.CSV

Task1.py

Task2.py

Task3.py

Task2.py

```
1 import pandas as pd
2 df=pd.read_csv("Data.CSV")
3 print("Dataset Shape:")
4 print(df.shape)
5 print("\nDataset Information:")
6 df.info()
7
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task2.py

Dataset Shape:

(4, 2)

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 4 entries, 0 to 3

Data columns (total 2 columns):

Column Non-Null Count Dtype

--- ---

0 Name 4 non-null str

1 Marks 4 non-null int64

dtypes: int64(1), str(1)

memory usage: 196.0 bytes

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

powerShell

powerShell

powerShell

OUTLINE

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EXPLORER

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Task4.py

Task4.py

Task1_DataCollection

Data.CSV

Task1.py

Task2.py

Task3.py

Task4.py

Task4.py

```
1 import pandas as pd
2 df=pd.read_csv("Data.CSV")
3 print("Rows Before Removing Duplicates:")
4 print(len(df))
5 df=df.drop_duplicates()
6 print("\nRows After Removing Duplicates:")
7 print(len(df))
8 print("\nDataset:")
9 print(df)
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task4.py

Rows Before Removing Duplicates:

4

Rows After Removing Duplicates:

4

Dataset:

	Name	Marks
0	Arun	85
1	Kumar	90
2	Priya	78
3	Divya	88

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

> OUTLINE

> TIMELINE

EXPLORER

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TASK1_DATACOLLECTION

Data.CSV

Task1.py

Task2.py

Task3.py

Task4.py

Task5.py

Task4.py

Task5.py

X

Task5.py

1 import pandas as pd

2 df=pd.read_csv("Data.CSV")

3 df["status"]="Pass"

4 print("Dataset After Transformation:")

5 print(df)

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task5.py

Dataset After Transformation:

	Name	Marks	Status
0	Arun	85	Pass
1	Kumar	90	Pass
2	Priya	78	Pass
3	Divya	88	Pass

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

> OUTLINE

> TIMELINE

EXPLORER

TASK1_DATACOLLECTION

Data.CSV

Task1.py

Task2.py

Task3.py

Task4.py

Task5.py

Task6.py

Task4.py

Task5.py

Task6.py

Task6.py

```
1 import pandas as pd
2 df=pd.read_csv("Data.csv")
3 print("Descriptive Statistics:")
4 print(df.describe())
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task6.py

Descriptive Statistics:

Marks

count	4.000000
mean	85.250000
std	5.251984
min	78.000000
25%	83.250000
50%	86.500000
75%	88.500000
max	90.000000

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

OUTLINE

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EXPLORER

✓ TASK1_DATACOLLECTION

- Data.CSV
- Task1.py
- Task2.py
- Task3.py
- Task4.py
- Task5.py
- Task6.py
- Task7.py

Task4.pyTask5.pyTask6.pyTask7.py X

Task7.py

```
1 import pandas as pd
2 df=pd.read_csv("Data.CSV")
3 filtered_data=df[df["Marks"]>80]
4 print("Students With Marks Greater Than 80:")
5 print(filtered_data)
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

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- PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task6.py

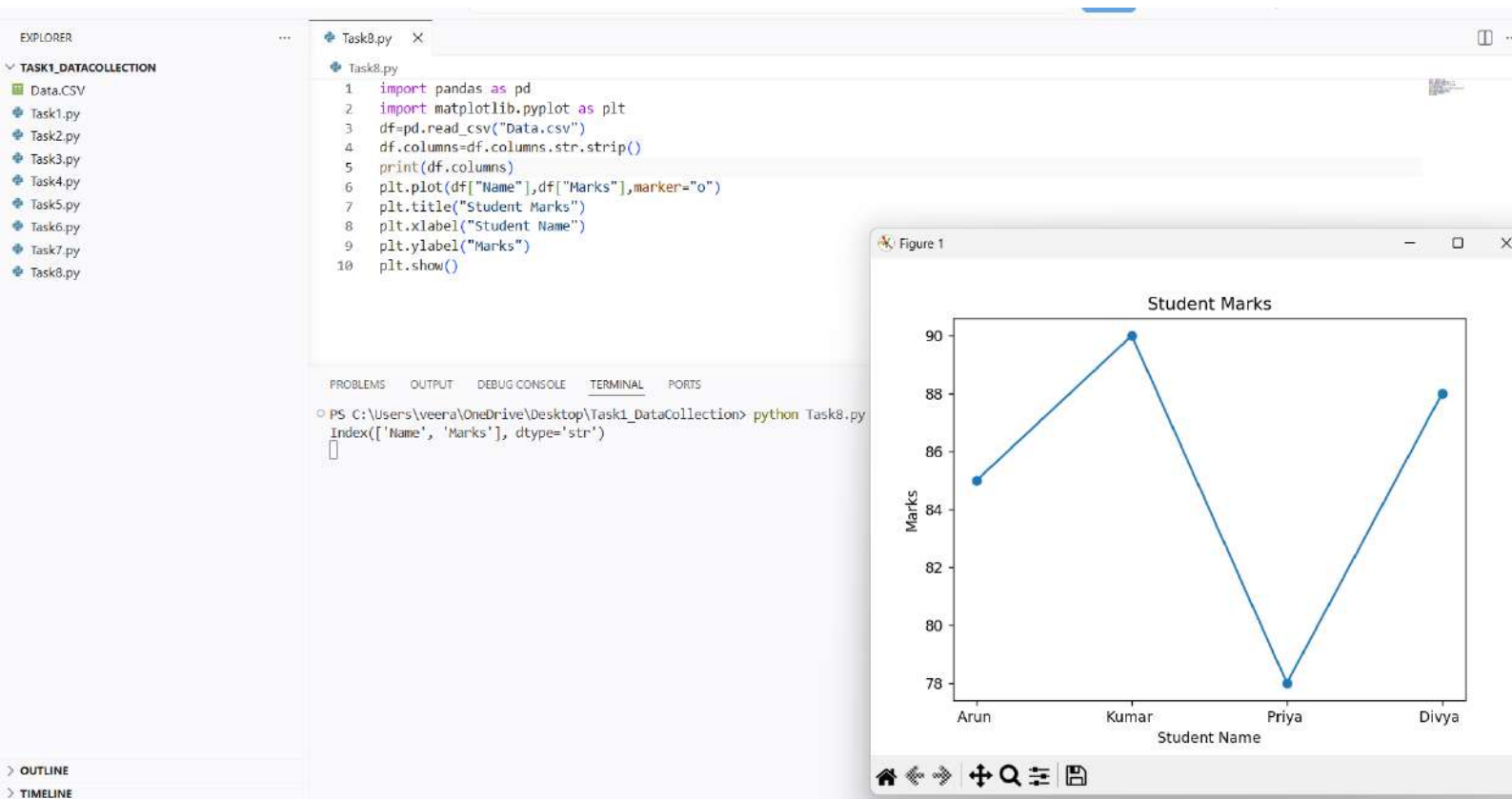
Descriptive Statistics:

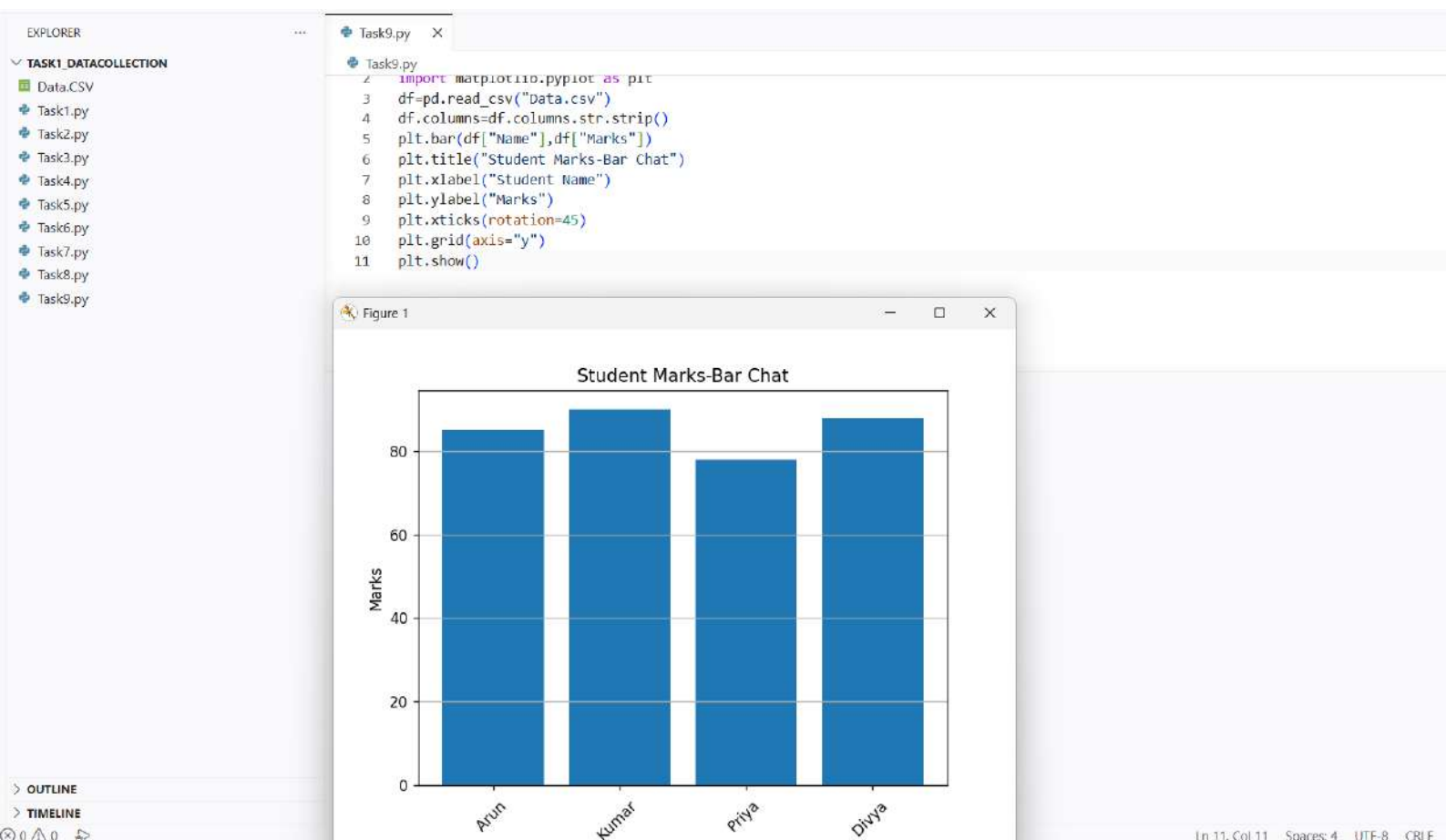
	Marks
count	4.000000
mean	85.250000
std	5.251984
min	78.000000
25%	83.250000
50%	86.500000
75%	88.500000
max	90.000000

- PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

OUTLINE

TIMELINE





EXPLORER

TASK1_DATACOLLECTION

- Data.CSV
- Task1.py
- Task2.py
- Task3.py
- Task4.py
- Task5.py
- Task6.py
- Task7.py
- Task8.py
- Task9.py
- Task10.py

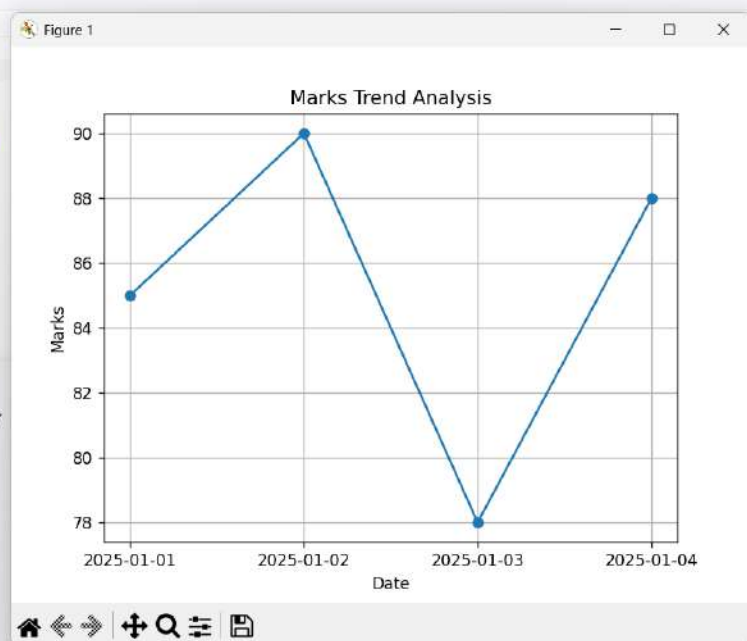
Task10.py

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 df = pd.read_csv("data.csv")
4 print(df.columns)
5 plt.plot(df["Date"], df["Marks"], marker="o")
6 plt.title("Marks Trend Analysis")
7 plt.xlabel("Date")
8 plt.ylabel("Marks")
9 plt.grid(True)
10 plt.show()
11
```

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PS C:\Users\veena\OneDrive\Desktop\Task1_DataCollection>
Index(['Date', 'Name', 'Marks'], dtype='str')

OUTLINE
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EXPLORER

▼ TASK1_DATACOLLECTION

Data.CSV

Task1.py

Task2.py

Task3.py

Task4.py

Task5.py

Task6.py

Task7.py

Task8.py

Task9.py

Task10.py

Task11.py

Task11.py

Task11.py

```
1 import pandas as pd
2 df = pd.read_csv("data.csv")
3 print(df.corr(numeric_only=True))
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

● PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task11.py

Marks

Marks 1.0

○ PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

> OUTLINE

> TIMELINE

EXPLORER

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TASK1_DATACOLLECTION

Data.CSV

Task1.py

Task2.py

Task3.py

Task4.py

Task5.py

Task6.py

Task7.py

Task8.py

Task9.py

Task10.py

Task11.py

Task12.py

Task12.py

Task12.py

```
1 import pandas as pd
2 df = pd.read_csv("data.csv")
3 result = df.groupby("Name")["Marks"].mean()
4 print(result)
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

● PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task12.py

Name

Arun 85.0

Divya 88.0

Kumar 90.0

Priya 78.0

Name: Marks, dtype: float64

○ PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

> OUTLINE

> TIMELINE

EXPLORER

TASK1_DATACOLLECTION

Data.CSV

Task1.py

Task2.py

Task3.py

Task4.py

Task5.py

Task6.py

Task7.py

Task8.py

Task9.py

Task10.py

Task11.py

Task12.py

Task13.py

Task13.py

```
1 import pandas as pd
2 from sklearn.linear_model import LinearRegression
3 df = pd.read_csv("data.csv")
4 df["Day"] = range(1, len(df) + 1)
5 X = df[["Day"]]
6 y = df["Marks"]
7 model = LinearRegression()
8 model.fit(X, y)
9 next_day = pd.DataFrame({"Day": [len(df) + 1]})
10 prediction = model.predict(next_day)
11 print("Predicted Marks:", prediction[0])
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python Task13.py

Predicted Marks: 84.5

PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection>

OUTLINE

TIMELINE

EXPLORER

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Task14.py X

Task14.py

```
1 import pandas as pd
2 df = pd.read_csv("data.csv")
3 df["Grade"] = df["Marks"].apply(
4     lambda x: "A" if x >= 80 else "B")
5 df.to_csv("processed_data.csv", index=False)
6 print("Data Exported Successfully")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\veena\OneDrive\Desktop\Task1_dataCollection> python Task14.py
Data Exported Successfully
- PS C:\Users\veena\OneDrive\Desktop\Task1_dataCollection> █

OUTLINE

TIMELINE

TASK1_DATACOLLECTION

- Data.CSV
- processed_data.csv
- Task1.py
- Task2.py
- Task3.py
- Task4.py
- Task5.py
- Task6.py
- Task7.py
- Task8.py
- Task9.py
- Task10.py
- Task11.py
- Task12.py
- Task13.py
- Task14.py

